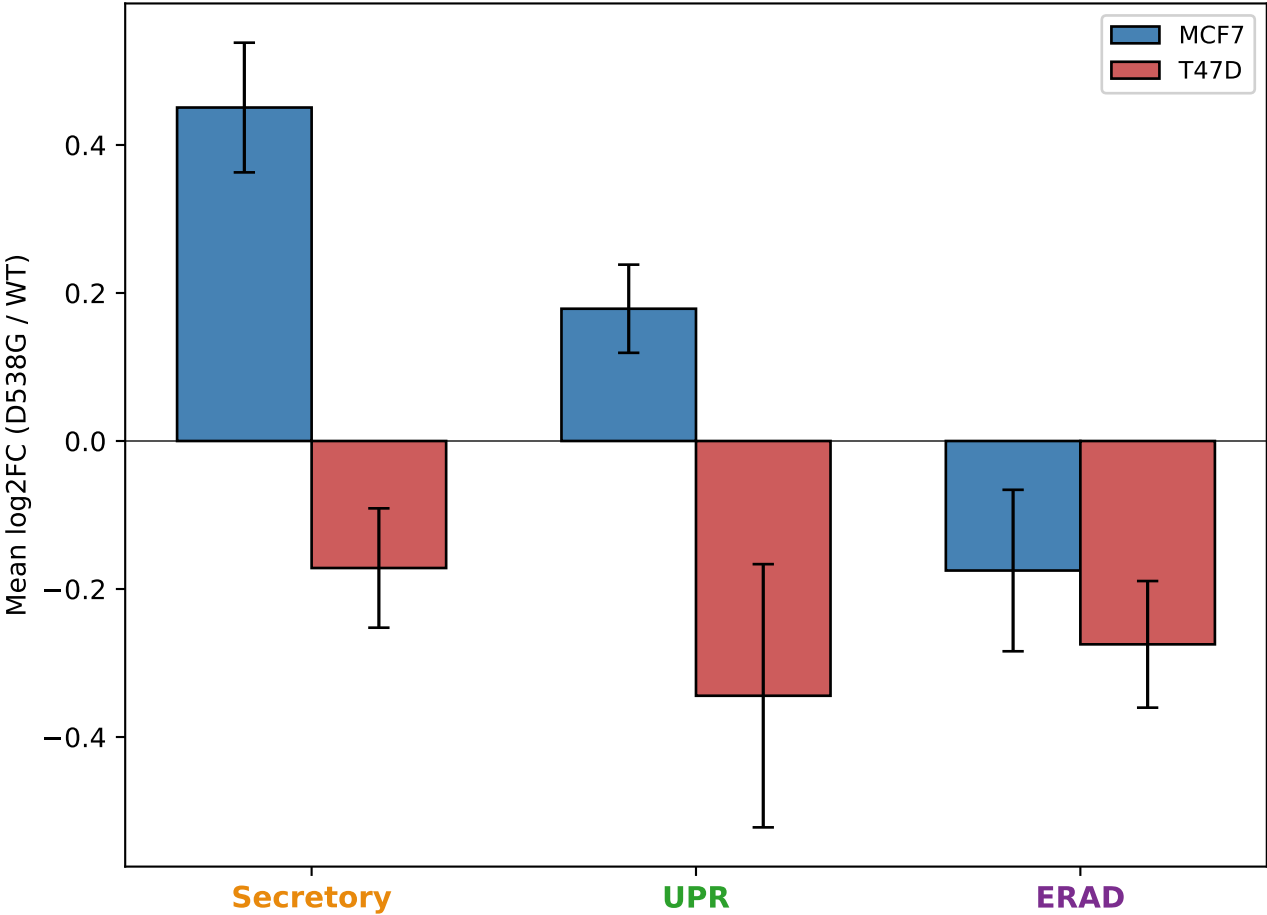


Functional Consequence: Divergent Proteostasis Determines MDK Fate

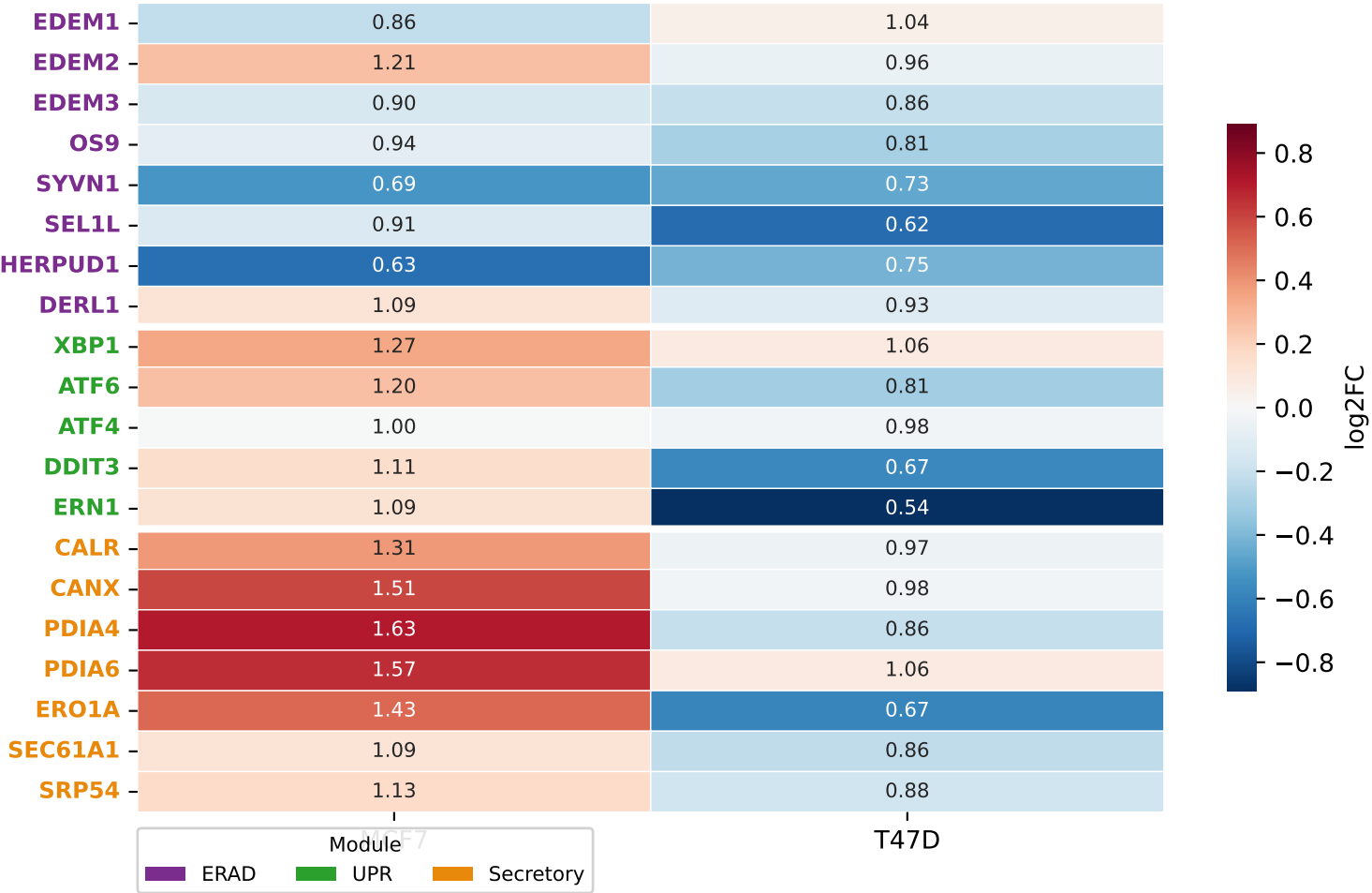
A

Module-Level Response



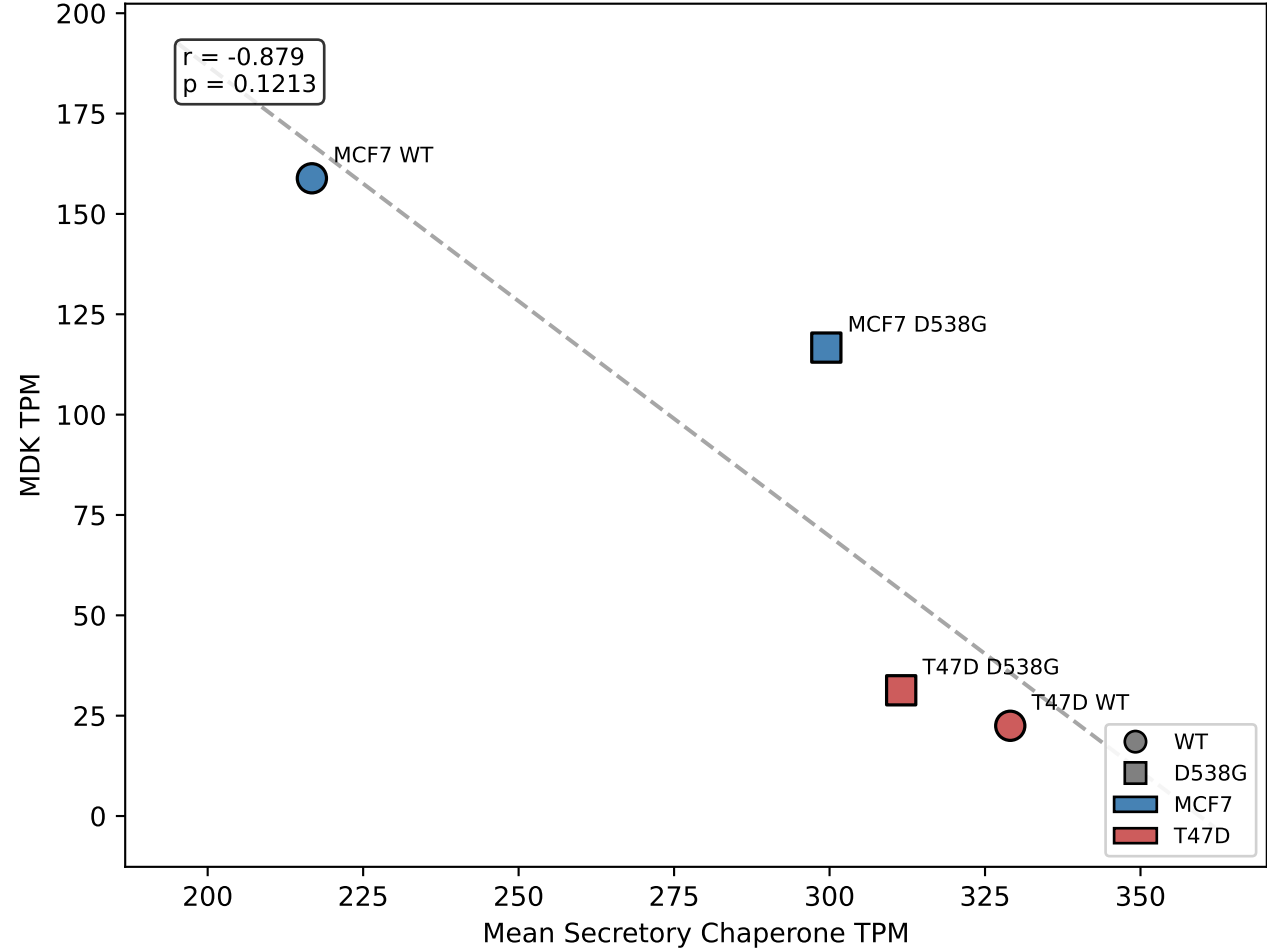
B

Gene-Level Fold Changes by Module



C

MDK mRNA vs Secretory Capacity



D

Integrated Evidence for Divergent Protein Fate

| Context                | Key Finding                                          | Dataset              | Statistic              | Experimental Prediction                             |
|------------------------|------------------------------------------------------|----------------------|------------------------|-----------------------------------------------------|
| Different cistrome     | 168/632 opposite genes have diff. ER binding         | GSE125117 + GSE89888 | $p < 0.05$ (Fisher)    | CHX chase: T47D-D538G shorter MDK half-life         |
| Different proteostasis | 93% secretory co-regulation, UPR activation (3/5 UP) | GSE89888             | $p=0.0009$ , $p=0.04$  | Pulse-chase: MCF7-D538G higher fractional secretion |
| Module coordination    | MDK mRNA anti-correlates with protein secretion      | GSE89888 + ELISA     | $p=0.0008$ , $p=0.002$ | Proteasome/lysosome inhibitors rescue T47D MDK      |